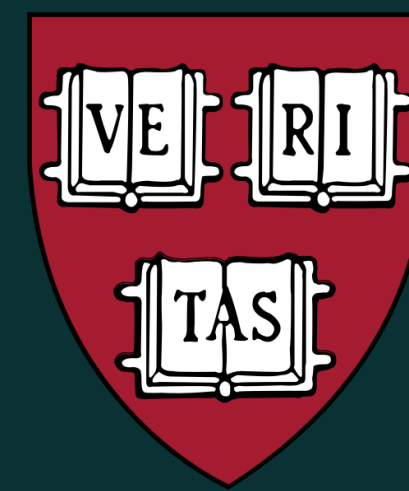


The Embedder's Dilemma: LLMs Are Better, but at What Cost?

Adnan El Assadi · Harvard

Niklas Muennighoff · Stanford

Jinhyuk Lee · Independent Researcher



Paper



Code

LLMs now match the best embedding models. At 1,431× the cost.

Summary

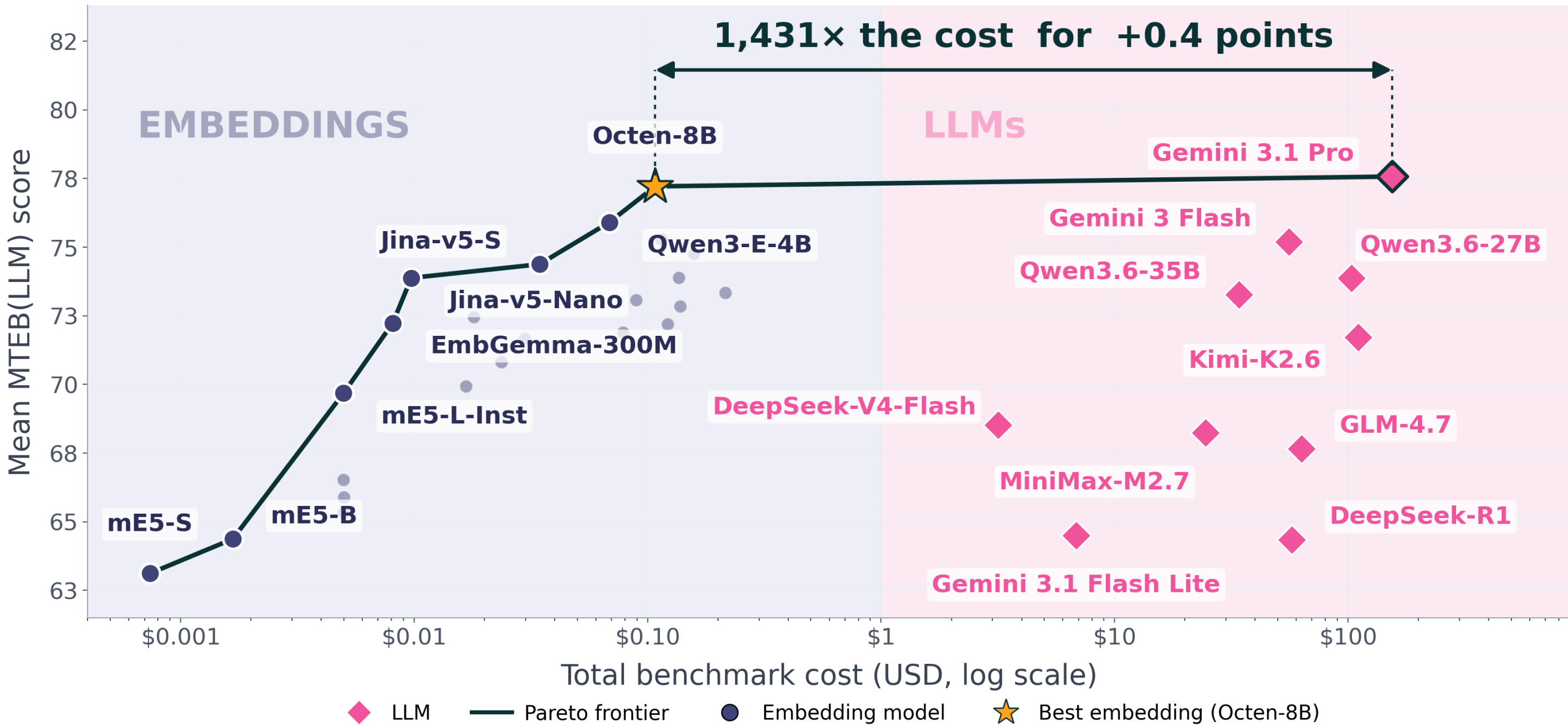
Should you replace your embedding pipeline with an LLM? We compare **10 LLMs** across six families with **26 embedding models** on **37 tasks** in five categories, with exact cost accounting for every model.

In aggregate they tie: 77.6 vs. 77.2. LLMs lead on reasoning-heavy retrieval; embeddings lead on classification; the other three categories are statistical ties.

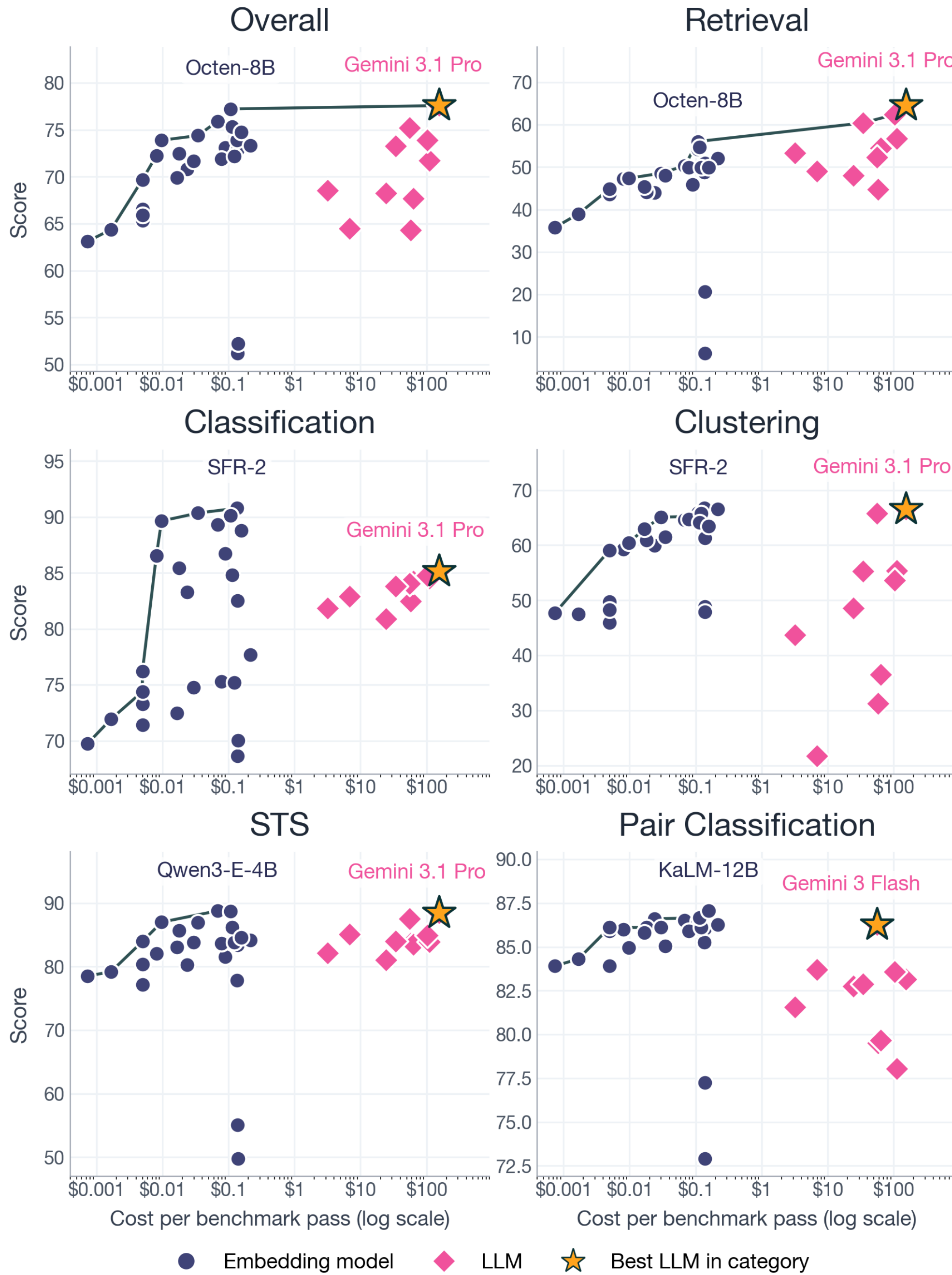
An LLM pays 1,431× more for that tie (\$154 vs. \$0.11 per benchmark pass), reasoning tokens are 28–81% of that bill, and embeddings run 736× faster on the same GPU.

Use embeddings by default, and an LLM only to rerank a shortlist on reasoning-heavy retrieval.

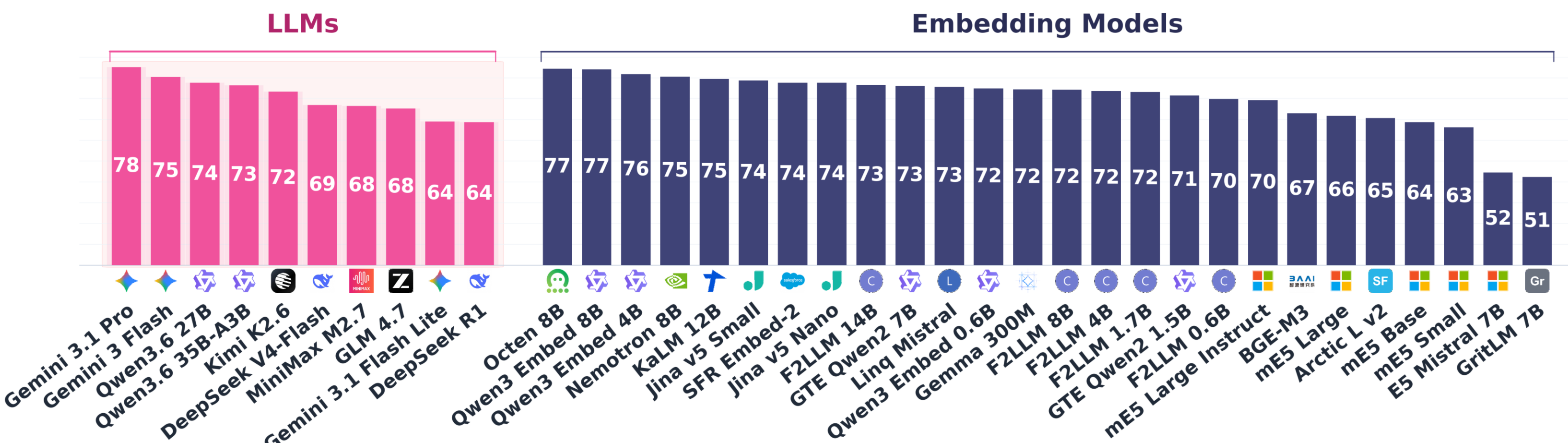
Cost vs. performance



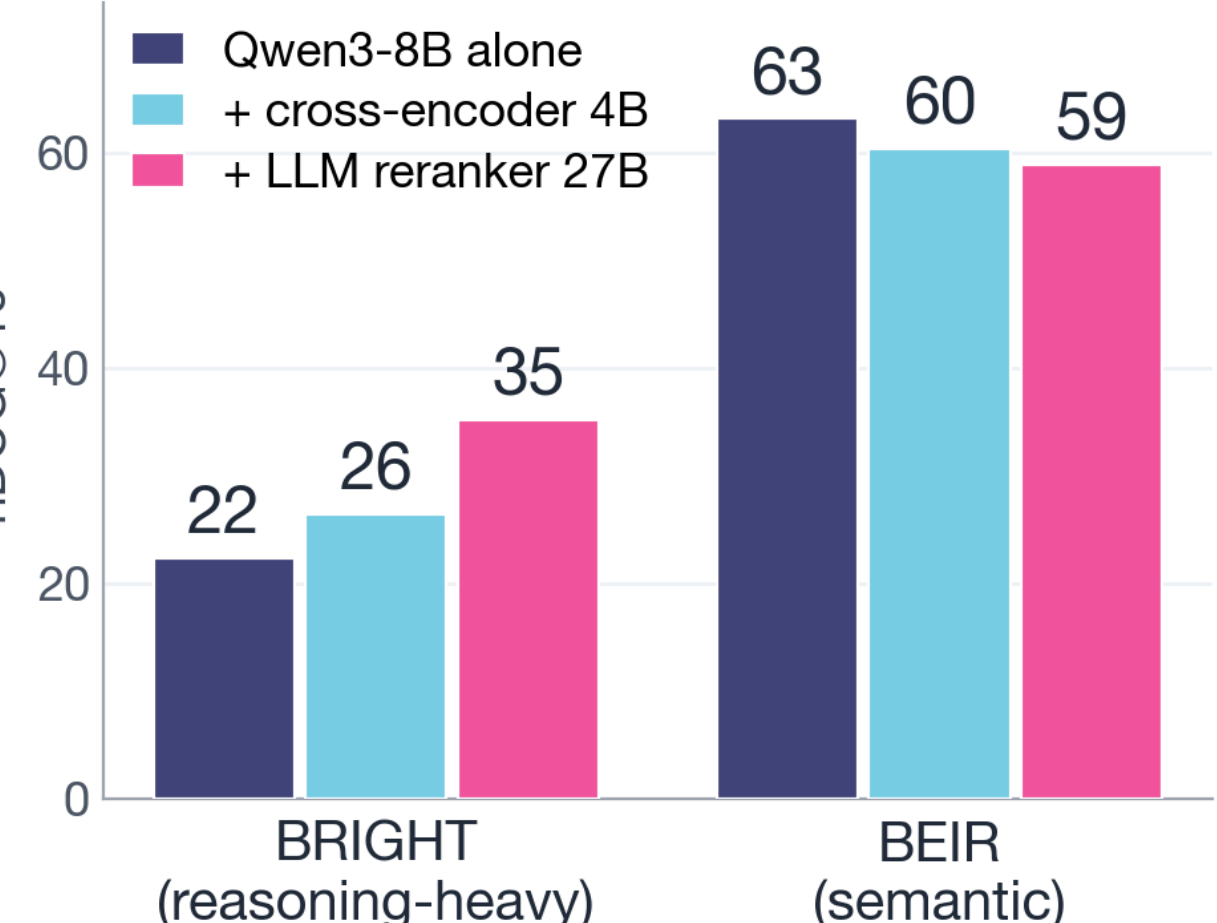
Cost vs. performance by category



Overall leaderboard



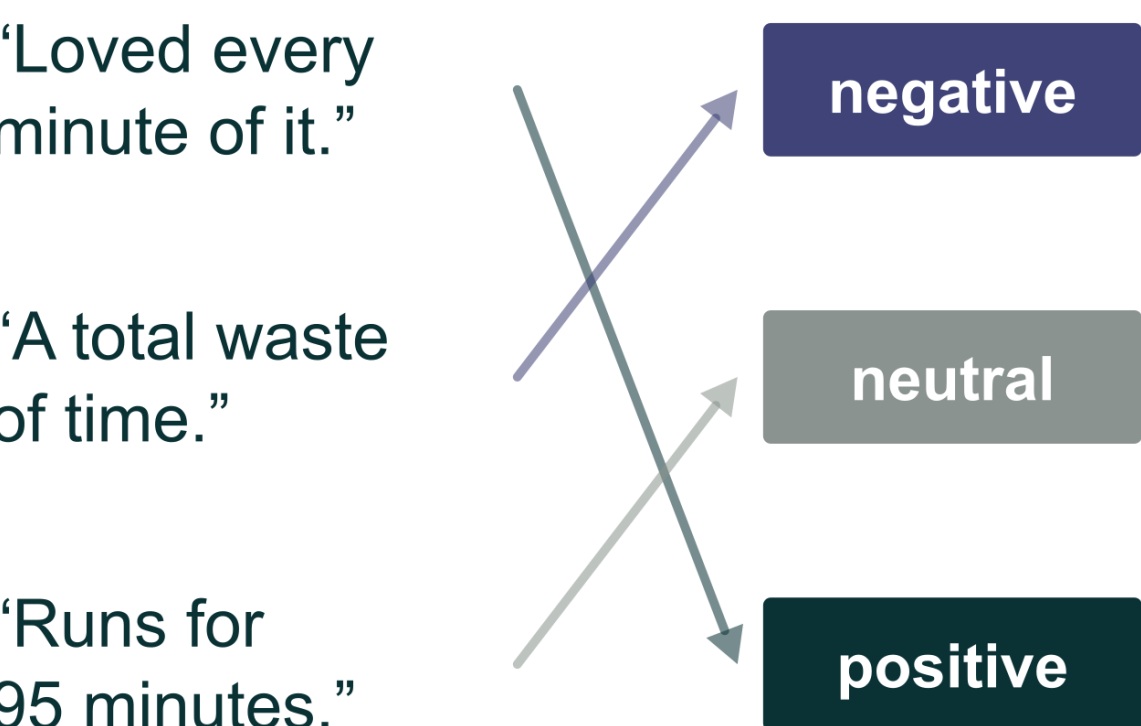
Reranking an embedding shortlist



MTEB(LLM): 37 tasks in five categories

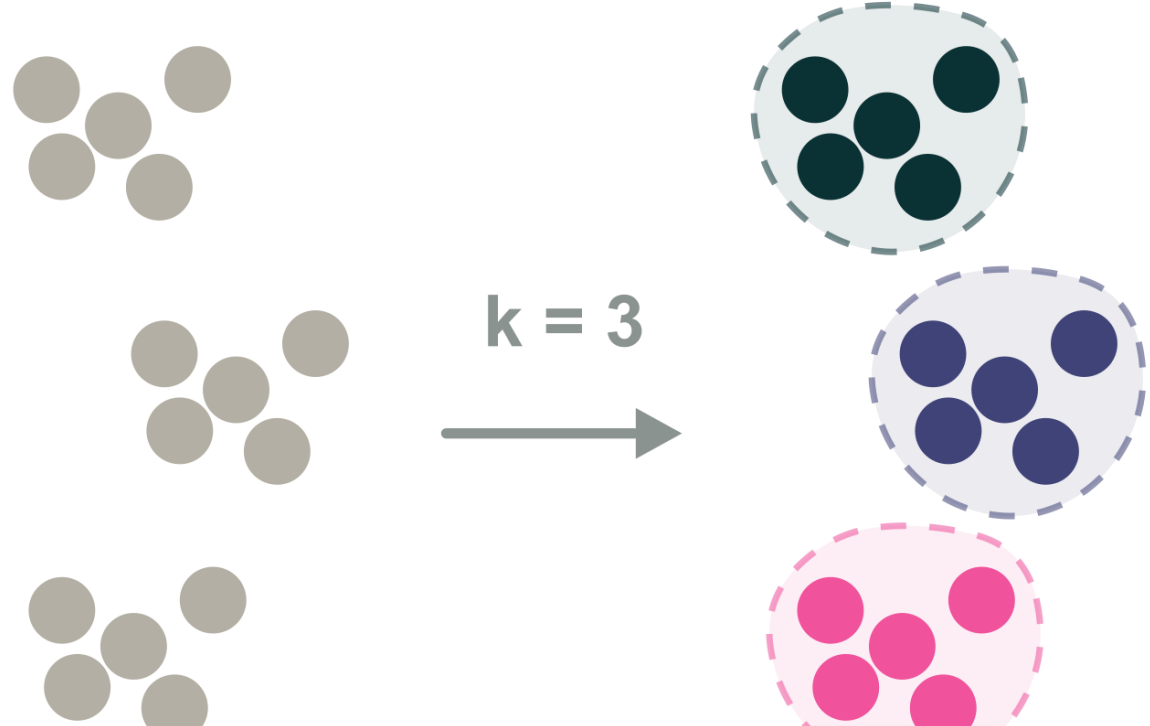
Classification

Which label fits?



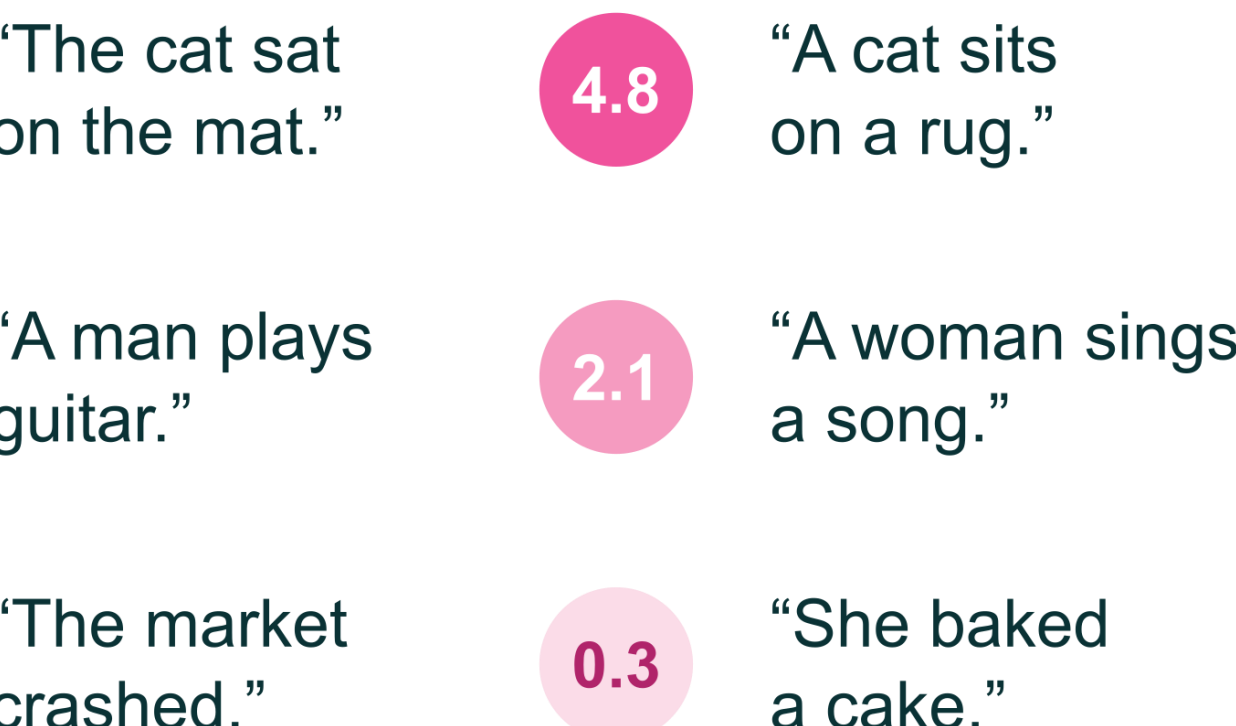
Clustering

Which belong together?



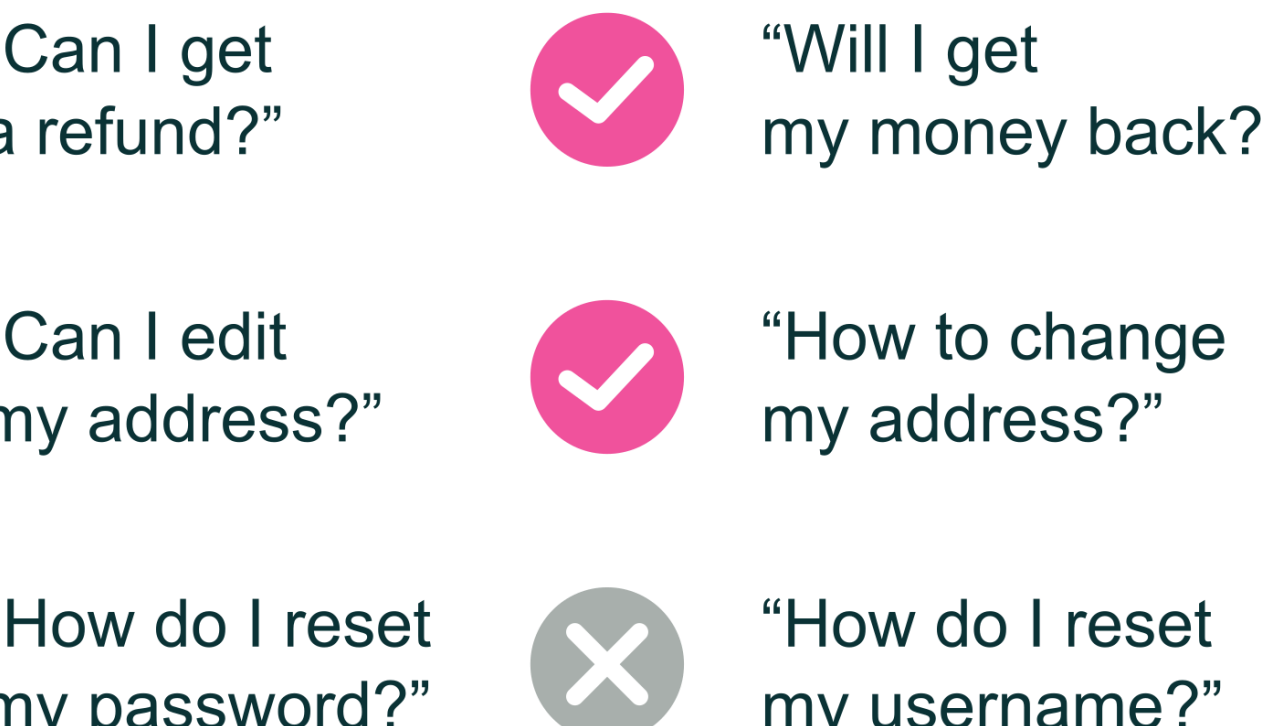
Semantic Similarity

How close in meaning?



Pair Classification

Do these ask the same thing?



Retrieval

Which document answers this?



Per task

